IDEA League

Master of Science in Applied Geophysics

Research Thesis

Test

Subtitle

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Test

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Master of Science Thesis

for the degree of Master of Science in Applied Geophysics

by

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JOINT MASTER'S IN APPLIED GEOPHYSICS

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Abstract

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Acronyms

DAS Distributed Acoustic Sensing

MPM Material Point Method

DEM Discrete Element Method

FE, FEM Finite Element Method

PST Propagation Saw Test

Chapter 1

Introduction

Snow avalanches are a major natural hazard in mountainous regions; they endanger human life as well as infrastructure (**schweizerSnowAvalancheFormation2003**). The Swiss Institute for Snow and Avalanche Research has reported an average of 22 fatalities caused by avalanches every year, with the number of non-fatal accidents being tenfold higher (**LongtermStatistics**). These data show that avalanches are a significhant threat, which is why it is crucial to understand how avalanches form and propagate in order to protect people and infrastructure better.

Avalanches are rapid masses of snow that move down mountain slopes due to gravity (**anceySnowAvalanches2001**). Snow avalanches can either be loose snow, which starts from a point in a cohesionless surface, wet snow, or snow slab avalanches (**schweizerSnowAvalancheFormation2003**). In this Master Thesis, the focus will be on dry snow slab avalanches, so mostly coherent masses of snow which contain no liquid water. Snow slab avalanches consist of a cohesive slab of snow (**schweizerSnowAvalancheFormation2003**) that is released because of loading at the surface, which triggers crack propagation in a weak layer (**gaumeSnowFractureRelation2017a**). Weak layers are characterized by the grain shape of snow, which leads to lower cohesion of those layers (**fohnMechanicalStructuralProperties1998**).

As mentioned, avalanches are a major threat to human life in mountainous regions, which is why detecting the breakoff of the snow slab is important for hazard mitigation. Detecting the breakoff of the slab means that warnings can be issued in lower parts of the mountain that an avalanche is possible. Furthermore, detecting slab breakoff with DAS is also important to further understand crack dynamics in snow as well as the factors influencing crack propagation.

In this master's thesis, the aim is to detect avalanche slab breakoff and crack propagation using modeled data from distributed acoustic sensing (DAS) cables.

In this thesis, a fibre-optic cable installed on the slope is treated as a continuous seismic sensor, so that deformation can be observed along its full length rather than at only a few isolated stations (**parkerDistributedAcousticSensing2014**). This is possible because DAS sends optical pulses through the fibre and records the backscattered light from every position along the cable (**shatalinInterferometricOpticalTimedomain1998**). Ground motion near the cable, for example weak-layer failure or slab breakoff, changes the fibre strain, which appears as phase variations in the backscattered signal (**hartogIntroductionDistributedOptical2017**). The phase variations are then converted into dynamic strain and from strain, into deformation along the cable (**trabattoniStrainDisplacementUsing2023**). mention also the cable location here, length etc

A pilot early-warning system using distributed acoustic sensing has been implemented in Norway, where avalanches are detected by their seismic signature (**turquetAutomatedSnowAvalanche2024**), meaning when the avalanche is already moving downhill. Such sensors are also located in Vallée de la Sionne in the Swiss Alps (**paitzPhenomenologyAvalancheRecordings2023**). The detectability of avalanches by DAS has also been studied in the Swiss Alps by **paitzPhenomenologyAvalancheRecordings2023** and **edmeFiberopticDetectionSnow2023**, but the detectability of the slab breakoff itself has not been studied.

A multitude of models are used to study different processes in snow. The SNOWPACK and CROCUS models used in the Swiss and French alps model the structure and evolution of snow (**brunNumericalModelSimulate1992**; **barteltPhysicalSNOWPACKModel2002**). These models were developed to understand how snow evolves under different conditions, for example a changing temperature gradient and water content and what consequences this has for the formation of weak layers where failure likely occurs and avalanches are triggered (**brunNumericalModelSimulate1992**; **barteltPhysicalSNOWPACKModel2002**). SNOWPACK and CROCUS are comprehensive models that take into account multiple governing equations and processes and what consequences they have for the structure of snow. While it is important to understand the influences on for example temperature gradient on the snow structure and weak layer formation and stability as a consequence, this is not the focus of the thesis. This is because these models are too complex and comprehensive for the purpose of trying to detect slab breakoff in DAS data as they model the entire snow structure. Using SNOWPACK or CROCUS to investigate how different snowpack compositions and factors influence the DAS signal is beyond the scope of this thesis.

To model crack propagation in snow, multiple numerical modeling methods have been used. **cundallDiscreteNumericalModel1979** developed the distinct element method (DEM) for granular media. This model discretizes the domain as spheres and disks, which interact with each other based on Newton's second law (**cundallDiscreteNumericalModel1979**). The interactions of particles are modeled by the governing equations and the resulting forces and displacements (**cundallDiscreteNumericalModel1979**). This method has been used by **bobillierMicromechanicalModelingSnow2020**; **bobillierNumericalInvestigationCrack2024**; **mulakNumericalInvestigationMixedmode2019** and many others to model crack propagation and its influencing factors. The main limitation of DEM is that the particle contacts and the corresponding equations need to be recalculated every time a grain contact is made or broken (**cundallDiscreteNumericalModel1979**). This means that for large, slope-scale simulations, this method becomes increasingly computationally expensive. In addition to the high computation cost, DEM also cannot account for anticrack propagation, the closure of cracks in this propagation mode is accounted for by manually removing elements from the mesh, which is not very efficient (**gaumeInvestigatingReleaseFlow2019**).

The same is true for finite element modelling (FE or FEM), which is done using SALVUS in this Thesis. FEM has been used to model snow mechanics since the 1970s, a review of the different approaches has been published by **podolskiyReviewFiniteelementModelling2013**. The finite element modelling approach used in this thesis is described more in detail in **??**. FE can also not account for anticrack propagation, similarly to DEM, mesh elements would need to be removed to account for anticrack propagation.

This problem is solved by using the material point method (MPM). In this method, the material is discretized in a set of Lagrangian points, which are used to track mass, deformation and momentum (**bardenhagenMaterialpointMethodGranular2000**). Because the points are not fixed, an Eulerian background mesh is needed to calculate spatial derivatives (**bardenhagenMaterialpointMethodGranular2000**). This method is not only more computationally efficient than DEM on the slope-scale, it can also account for anticrack behavior and strain softening of snow (**gaumeInvestigatingReleaseFlow2019**). This is why the material point method will also be used in the thesis to investigate slab breakoff in DAS data on a large scale. The material point method has been used by **bergfeldSignaturesSubRayleighSupershear2025**, who modeled slope-scale experiments to understand the transition of crack propagation modes, **trottetTransitionSubRayleighAnticrack2022** who modeled propagation saw tests to observe transitions in crack propagation speeds, as well as **gaumeInvestigatingReleaseFlow2019** and **gaumeDynamicAnticrackPropagation2018**.

While MPM is very good at accounting for anticrack propagation, FE is more computationally efficient and accurate **lianAdaptiveFiniteElement2012**. This is relevant especially for large-scale simulations, such as the ones done in this Thesis. This is because in MPM, the discretization through particles causes the displacement of each particle to be tracked and the contact at each particle to be calculated (**zhangModellingLargescaleLandslide2023**). Furthermore, MPM does not include boundary conditions in the way FE does (**coombsImmersedFEMMPM2025**). Finite element is more efficient on the slope scale because the mesh is fixed, as described by (**FiniteElementMethod**). This means that the degrees of freedom and thus the number of calculations which need to be done are much smaller than in the MPM case. This means that MPM is more accurate at representing snow as a medium with its porous character and anticrack behavior, but FE is more efficient at performing large-scale simulations.

While it is important to understand the different modeling methods and their limitations, it is also important to understand the different types of crack propagation and what factors influence them. Understanding how fast a crack can propagate and which factors influence this is important for warning, as it relates the time of the initial crack to the time where the slab breaks off depending on snow and terrain parameters.

Two main types of crack propagation are believed to be important for weak layer failure: anticrack propagation and shear crack propagation (**bergfeldSignaturesSubRayleighSupershear2025**). There are three ways in which cracks can open up: mode I, where space is opened between the crack faces under tensile loading, and modes II and III, which represent parallel shearing and diagonal shearing (**heierliAnticracksNewTheory2003**). In anticrack propagation, the weak layer consolidates under loading (**knightSuperheatedIceTrue1972**), while cracks propagate in a combination of modes I-III (**heierliAnticracksNewTheory2003**). **gaumeDynamicAnticrackPropagation2018** modeled anticrack propagation induced by a propagation saw test (PST) using MPM, which showed that the scale of the simulations influences the propagation direction of the cracks. For smaller scale simulations, cracks propagate from top to bottom of a slope, whereas for slope-scale simulations, cracks propagate bottom to top, which means that remote triggering of avalanches is possible (**gaumeDynamicAnticrackPropagation2018**). This is an important consideration for the thesis because it influences where the slab breakoff would be observed in terms of DAS cable location. Other models also show that the crack propagation speed of anticracks is influenced by slope angle and curvature: concave slope areas have higher propagation speeds than convex areas of the slope (**gaumeInvestigatingReleaseFlow2019**). Furthermore, there is a directionality in propagation speed, simulations by **gaumeInvestigatingReleaseFlow2019** show that cracks propagate twice as fast upslope than in the downslope direction, which is especially important when thinking about remote triggering. The slab breaks off when the slope angle exceeds the friction angle of the material (**gaumeInvestigatingReleaseFlow2019**). Slope angle also influences which mode of crack propagation is more dominant (**rosendahlModelingSnowSlab2020**). In steeper terrains, shear modes (II and III) dominate, whereas in flatter terrain, mode I dominates the crack propagation (**rosendahlModelingSnowSlab2020**). The critical force, which is for example exerted by a skier, needed for weak layer failure is also dependent on slope angle; it decreases with increasing slope angle (**rosendahlModelingSnowSlab2020**). Snow parameters also have an influence on this critical force, as shown by simulations by **rosendahlModelingSnowSlab2020**: the thicker the slab or the higher the slab tensile strength, the better the force by the skier can be distributed on the weak layer and the more force is needed for failure.

Multiple experiments and numerical models have shown that cracks can propagate faster than the elastic wave speed in snow, so-called supershear crack propagation (**trottetTransitionSubRayleighAnticrack2022**; **bobillierNumericalInvestigationCrack2024**; **bergfeldSignaturesSubRayleighSupershear2025**). Understanding what causes the transition from sub-Rayleigh to supershear crack propagation is of high importance for the thesis, because it can give an estimate of whether supershear crack propagation will occur under certain conditions, which leads to faster propagation of cracks and less time between loading and slab breakoff. In MPM models of PST, **trottetTransitionSubRayleighAnticrack2022** observe that the occurrence of supershear cracks is dependent on slope angle: in steeper terrains, supershear propagation was observed. This can be explained by slab bending, because in steep terrains, bending of the slab is not limited to the collapse depth of the weak layer (**trottetTransitionSubRayleighAnticrack2022**). In in steeper terrain, the slab can bend more because of the terrain angle, the additional bending increases tensile stress in the slab, which leads to faster crack propagation (**trottetTransitionSubRayleighAnticrack2022**).These observations were also made by **bobillierNumericalInvestigationCrack2024** as well as

bergfeldSignaturesSubRayleighSupershear2025. The release area of an avalanche is also tied to propagation speed: at supershear propagation speed, there is less crack arrest and therefore larger release areas (**trottetTransitionSubRayleighAnticrack2022**; **bergfeldSignaturesSubRayleighSupershear2025**). This is crucial because larger release areas also mean a larger snow volume, which has more potential for damage.

Independent of crack propagation mode, different factors influence the speed and distance at which cracks can propagate in the weak layer before failure occurs (**gaumeDynamicAnticrackPropagation2018**; **gaumeSnowFractureRelation2017a**; **gaumeModelingCrackPropagation2015**). As discussed above, slope angle and stiffness are important factors. Other factors include weak layer parameters: the weaker the weak layer, the slower crack propagation; the thicker the weak layer, the lower the crack propagation distance before failure (**gaumeModelingCrackPropagation2015**). The depth of the slab also has an influence on propagation distance and speed: the thicker the slab, the larger the propagation distance and velocity (**gaumeModelingCrackPropagation2015**). How fast a slab will break off is also dependent on the critical crack length needed for failure (**gaumeSnowFractureRelation2017a**). If the snowpack is more stable, which is the case when the slab has a high elastic modulus or the weak layer is thicker, the critical crack length is higher (**gaumeSnowFractureRelation2017a**) because the snow can sustain longer cracks before it fails. A very important factor in material strength is loading rate: if loading is slow, sintering can occur (**mulakNumericalInvestigationMixedmode2019**). This means that new snow contacts can form, which can strengthen the material (**mulakNumericalInvestigationMixedmode2019**). This is in contrast to rapid loading, which happens for example due to a skier (**mulakNumericalInvestigationMixedmode2019**).

In order to understand how crack propagation and slab breakoff would look in DAS data, numerical modeling of the wave propagation is done. This is done in SALVUS, which is a package that uses spectral elements to model seismic wavefields (**afanasievSalvusFlexibleHighperformance2017**).

The different snow processes described have differenit implications for the seismic wavefield, which will be modeled. Firstly, the loading which occurs before crack nucleation changes the stress state of the snowpack. The weak layer collapse and anticrack propagation produce volumetric compaction in the snowpack, this produces a seismic signal. The crack propagation, whether sub-Rayleigh or supershear, creates a moving rupture front, which can be likened to a moving source of seismic waves, the cracks itself also create a seismic signal. Lastly, the slab breakoff and sliding creates a seismic signal because a large mass of snow moves downhill, which likely creates the largest seimsic signal of all of the processes mentioned. The signals of the different processes described will be modeled in the thesis, which will help to understand how the different processes look in DAS data and whether they can be distinguished from each other.

Based on the above information, there are multiple considerations necessary in order to model DAS data of crack propagation accurately. The scale of the models is an important factor, because as mentioned, it influences the direction of crack propagation. This influences at which location along or relative to the DAS cable slab breakoff will occur, which changes the measured data. Furthermore, multiple parameters influence the time passed between initial cracking and slab breakoff, these are important to consider when interpreting data.

The different modes of crack propagation are also important to consider, it is possible that a transition between sub-Rayleigh and supershear crack propagation can be observed in the data. The loading rate can also be considered, because slower loading leads to stabilizing processes such as sintering.

This thesis will focus on the modelling of crack propagation, as recorded by DAS. Sub-Rayleigh as well as supershear crack propagation will be modelled using different methods and models. The aim is to determine what the different cases of crack propagation will look like in DAS data. This is then relevant for detecting actual avalanches in real-world data. The main questions are what crack propagation in different regimes looks like, how it differs when different modeling methods are used, and how the results can be explained in terms of wave physics.

Part I

First Part

Chapter 2

First Real Chapter

This is a demonstration chapter. I will explain some of the possibilities of LaTeX. Here something will be shown of control theory, 'the transfer function' $H(s)$. Subscripts and superscripts can be put in the nomenclature list. Other things can also be added to the nomenclature list, like explanations of symbols being used throughout the thesis.

2-1 First section

This is the section. Referring to equations, figures and tables can easily be done by the commands `\eqnref{}`, `\figref{}` and `\tabref{}`.

$$H(s) = \frac{1}{s+2} \tag{2-1}$$

You see? Refer to equations like this Eq. (**??**), i.e. the name of the label you have given the specific equation, figure or table.

2-1-1 The first subsection

Now I demonstrate, numbering equations, using subequations:

$$\nabla \times \mathbf{L} = \frac{\partial \mathbf{G}}{\partial t} \tag{2-2a}$$

$$\nabla \times \mathbf{G} = \frac{\partial \mathbf{L}}{\partial t} + \mathbf{J} \tag{2-2b}$$

$$\mathbf{G} = \sigma \mathbf{J} \tag{2-2c}$$

Or we can make matrices:

$$\mathbf{Q}_{12} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

This can also be done using the `\align{}` command. Equation arrays are also possible:

$$\nabla \times \mathbf{L} = \frac{\partial \mathbf{G}}{\partial t} \tag{2-3}$$

$$\nabla \times \mathbf{G} = \frac{\partial \mathbf{L}}{\partial t} + \mathbf{J} \tag{2-4}$$

$$\mathbf{G} = \sigma \mathbf{J} \tag{2-5}$$

The first sub-subsection with a very very very long title, but in the table of contents one can only see the short title in square brackets

Impressed by the capabilities? If you want to know more about the capabilities of LaTeX, take a look at the "**The Not So Short Introduction to LaTeX 2_ε**", which can be found on the internet.

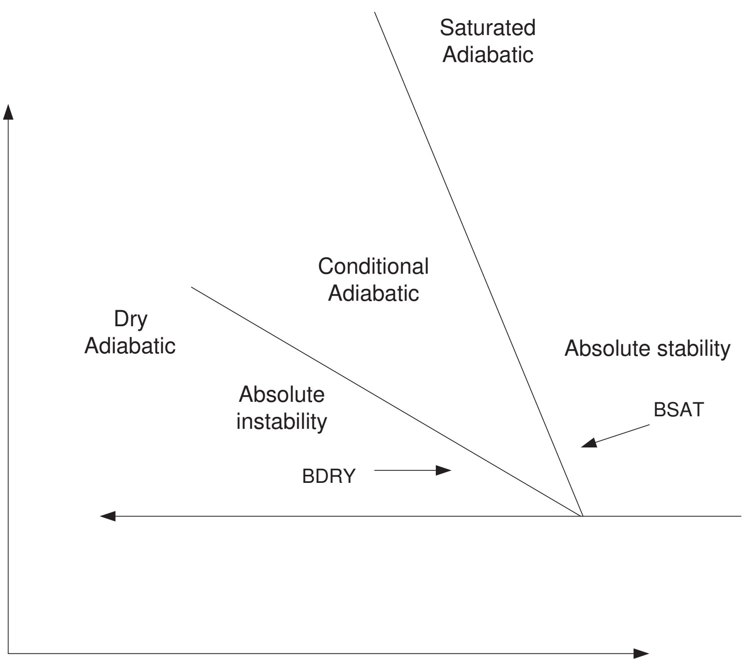

Figure 2-1: Stability conditions for the vertical stability of saturated and unsaturated air.

Next paragraph. .
And finally I end this example file with a table which will be centered in the middle of the following page

Data **files listed in a table**	
First part	
Fabracadabra.m	- Saturation computation
Fobracadabra.m	- Pressure computation
Fibricadibri.m	- Permeability computation
Structural rock model	
struct.m	- Rock structural data using symmetric boundary condition
bstruct.m	- Rock structural data using anti-symmetric boundary condition

Table 2-1: Good-looking program data deck files.

Appendix A

The back of the thesis

A-1 An appendix section

A-1-1 An appendix subsection with C++ Lisitng

A-1-2 A Matlab Listing

Appendix B

Yet another appendix

B-1 Another test section

Ok, all is well.